

EE1060: Differential Equations and Transform Techniques**Solutions to Practice Problem Set - 02**

$$1. i(t) = \frac{V_{dc}}{R} \left(1 - e^{-\frac{R}{L}t} \right) + I_0 e^{-\frac{R}{L}t}$$

$$2. y(x) = \frac{A(k \cos(\omega x) + \omega \sin(\omega x))}{k^2 + \omega^2} + C e^{-kx}$$

$$3. \frac{y^3}{3} + x^3 + y + 2x = c$$

$$4. y^3 + 2x^2y + 4xy = C$$

$$5. \frac{x^2}{y^2} + x - \frac{1}{y} = C$$

$$6.$$

$$7. \frac{1}{ce^{ax} + \frac{b}{a}}$$

$$8. y = cx^4 - x^2$$

$$9. y = \frac{x \ln x}{2} - \frac{x}{4} + \frac{c}{x}$$

$$10. y(t) = 1000 \left(1 - \frac{1}{1 + 0.25e^{0.05t}} \right)$$

$$11. x^3 - 4xy^2 + c = 0$$

$$12. n = 1$$

$$13. x^3y^2 - 6xy = C$$

$$14. y^2x + e^y(2y - y^2) = c$$

$$15. y = \frac{1}{cx - \frac{x^3}{2}}.$$

$$16. y = ce^{x + \frac{x^2}{2}} - 1$$

$$17. y = \frac{3[(1+x)\sin x + \cos x] - 2}{(1+x)^2}$$

$$18. y = \frac{1}{2}\sqrt{x^2 + 4} - \frac{4}{(x^2 + 4)^{\frac{3}{2}}}$$

$$19. y = \frac{1}{(x^2 + cx)}$$

$$20. y = \frac{-e^{-x}}{(x + c)}$$

$$21. y = x \left(1 + \frac{1}{(\ln(c\sqrt{x}))} \right)$$

22. $y = ce^{-x^3} + 2$

23. $\left(\frac{e^{3x}}{e^{3x} + x^{3x}}\right)^{1/3}$

24. $i(t) = \frac{V_0}{(R - L)} (e^{-t} - e^{-tR/L})$

References

- [1] G. F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd ed. Boca Raton, FL: CRC Press, 2016.
- [2] C. H. Edwards and D. E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th ed. Upper Saddle River, NJ: Pearson, 2000.